

No-three-collinear lattice walks: the joint minimum problem

A sharp-threshold follow-up to Erdős Problem 193

Definitions. A finite or infinite lattice walk (X_n) is *admissible* if $X_0 = 0$, its vertices are pairwise distinct, and

$$\text{rank}_{\mathbb{Q}}[X_j - X_i \quad X_k - X_j] = 2 \quad \text{for every } 0 \leq i < j < k$$

within its index range. Thus no three vertices lie on an affine line. For $d, s \geq 1$, define

$$\begin{aligned} \mathcal{B}(d) &\iff \exists \text{ an infinite admissible } (P_n) \subset \mathbb{Z}^d : \quad P_{n+1} - P_n \in \{e_1, \dots, e_d\} \quad (n \geq 0), \\ \mathcal{E}(3, s) &\iff \exists S \subset \mathbb{Z}^3 \setminus \{0\}, \quad |S| \leq s, \quad \exists \text{ an infinite admissible } (Q_n) \subset \mathbb{Z}^3 : \quad Q_{n+1} - Q_n \in S \quad (n \geq 0), \\ d_* &= \min\{d \geq 1 : \mathcal{B}(d)\}, \quad s_* = \min\{s \geq 1 : \mathcal{E}(3, s)\}. \end{aligned}$$

Here e_r is the r th positive standard basis vector. In $\mathcal{E}(3, s)$, S is chosen *once for the entire infinite walk*; its vectors are arbitrary nonzero integer vectors, with no prescribed coordinate bound. Both minima are finite by the original finite-step construction and basis encoding.

Problem. Determine the exact ordered pair (d_*, s_*) : prove unconditional infinite existence at each claimed minimum and impossibility for *every* walk with a smaller dimension or step budget, respectively. In particular, determine whether the compression gap $s_* - d_*$ is zero. Equality alone would not determine the common value.

Basic reduction and bounds. Label the vectors of a successful step set by $1, \dots, t, t \leq s$, and replace each step of type r by e_r . The resulting count walk projects linearly onto the original walk, so is itself admissible. Hence

$$\mathcal{E}(3, s) \implies \mathcal{B}(s), \quad \boxed{4 \leq d_* \leq s_* \leq 16}.$$

The lower bound follows because every ternary word of length eight contains an abelian square; the ceiling comes from the original sixteen-step 3D construction. The converse implication $\mathcal{B}(s) \implies \mathcal{E}(3, s)$ is *not assumed*. Equivalently, d_* is the minimum number of step types when the ambient lattice dimension is unrestricted. A written six-step construction proposes $s_* \leq 6$, hence $d_* \leq 6$, but independent review remains pending. Subject to that verification, the candidate pairs are $(4, 4), (4, 5), (4, 6), (5, 5), (5, 6), (6, 6)$; none is asserted as the answer.

Equivalent word and projection formulations. Let $\mathcal{W}_d$ consist of infinite words on $\{1, \dots, d\}$ having no adjacent nonempty blocks x, y with $\psi(x)/|x| = \psi(y)/|y|$, where ψ is the Parikh (letter-count) vector. The lengths $|x|, |y|$ may differ: avoiding ordinary abelian squares alone is insufficient. If $P_n(w)$ counts the first n letters, then

$$\mathcal{B}(d) \iff \mathcal{W}_d \neq \emptyset, \quad \mathcal{E}(3, d) \iff \exists w \in \mathcal{W}_d \exists A \in \mathbb{Z}^{3 \times d} \forall i < j < k : \text{rank}_{\mathbb{Q}}[A(P_j - P_i) \quad A(P_k - P_j)] = 2.$$

Thus $s_* = d_*$ requires a fixed integer projection for *one* optimal-alphabet witness, not for every valid word. For basis walks only, $\sum_r (P_n)_r = n$ makes noncollinearity equivalent to $(k - j)(P_j - P_i) \neq (j - i)(P_k - P_j)$. This time-weighted test is not sufficient for arbitrary 3D step sets.

Equivalent quantifier distinction. Let $\mathcal{F}_N(S)$ mean that an admissible N -step walk in $\mathbb{Z}^3$ uses steps from S . In the following formulas, menus are finite subsets of $\mathbb{Z}^3 \setminus \{0\}$:

$$\boxed{\begin{aligned} \mathcal{B}(s) &\iff \forall N \geq 1 \exists S_N, \quad |S_N| \leq s : \mathcal{F}_N(S_N), \\ \mathcal{E}(3, s) &\iff \exists S, \quad |S| \leq s \forall N \geq 1 : \mathcal{F}_N(S). \end{aligned}}$$

These follow from integer projection of finite basis prefixes, basis encoding, and König's lemma. A uniform bound on all menu coordinates in the first line suffices to obtain the second; unbounded prefix-dependent projections do not. Finite positive evidence and lower bounds confined to one construction family do not settle either minimum.